

CHEMISTRY EXAM-25: DETAILED MARKING REPORT

Chapter 10 (Acids, Bases & Salts, 70%) + Revision: Chapters 6 and 8 (30%)

Seemab · Class 9 · Attempt Date: 2026-09-22 · Paper: 50 marks, 2h15m, no optional questions · Second Chapter 10 paper

Final Score: 46.5 / 50 · 93.0% · Grade A

Deep check 2026-09-22 corrected the first pass from 47 to 46.5 (Q.9(ii) missing unit). See the judgement-calls table for the full range.

**Fairness context checked (per Atif's note).** At ~14:45 on 2026-09-22 Seemab received Chapter 10 Aasan Notes v1, which wrongly said "H3O+ is never an acid"; this was corrected in v2 at ~15:40. Every appearance of H3O+ on this paper (Q.4i, Q.8i-a) is as a **product**, and she correctly named the proton donor on the **left** of the arrow as the acid in every case: the same correct habit that would apply with or without the v1 wording, so **no lost mark traces to the wrong v1 trick**. Separately, her textbook's own worked examples print three neutralization equations unbalanced (Na2CO3+HCl, H2SO4+KOH, H2SO4+NaOH); none of the three appears in her Q.5 answers (she answered NH4OH+HNO3 and H2SO4+Mg(OH)2, both correctly balanced), so this is **not applicable** to this attempt. The Chapter 9 contact-process catalyst does not appear anywhere on this paper.

10 / 10  
SECTION A (MCQS)

21 / 24  
SECTION B (SHORT)

15.5 / 16  
SECTION C (LONG)

NEW CHAPTER (Ch 10): A Q1–7, B Q2–6, C Q8  
35 / 35 (100%)

REVISION (Ch 6, 8): A Q8–10, B Q7, C Q9  
11.5 / 15 (76.7%)

SECTION A: Multiple Choice Questions (10 / 10)

| #  | Question (summary)                        | Correct                   | Seemab's | Result          |
|----|-------------------------------------------|---------------------------|----------|-----------------|
| 1  | Cannot be classified as an Arrhenius acid | C: CO2                    | C        | 1               |
| 2  | A Bronsted-Lowry base                     | A: NH3                    | A        | 1 RE-TEST FIXED |
| 3  | Salt formed: Mg(OH)2 + HCl                | C: MgCl2                  | C        | 1               |
| 4  | Ammonia is a base because it              | D: accepts a proton       | D        | 1 RE-TEST FIXED |
| 5  | A weak base                               | B: Al(OH)3                | B        | 1               |
| 6  | Property acids and bases share            | D: conduct electricity    | D        | 1               |
| 7  | Definition of a strong base               | A: dissociates completely | A        | 1               |
| 8  | [Revision] ΔH°=+131.4kJ reaction is       | B: endothermic            | B        | 1 RE-TEST FIXED |
| 9  | [Revision] Mass of 0.75 mol CO2           | B: 33 g                   | B        | 1               |
| 10 | [Revision] Moles in 54 g of water         | A: 3 mol                  | A        | 1               |

Perfect 10/10: every option independently re-derived from the exam paper (not just checked against the key) and cross-checked against her handwritten list on the last page. No misreads, no key errors found. This is her best-ever MCQ result on a Chemistry paper of this size.

SECTION B: Short Questions (21 / 24)

Q.2: Arrhenius acids, strong acid, weak base 4 / 4

- (i) "An acid is a substance that ionizes in water to produce H+ ions." Examples: "HCL → H+ + Cl- ", "H2S04 → 2H+ + S04^-2"
- (ii) "HCl is a strong acid because it completely ionizes in water, producing a high concentration of H+ ions."
- (iii) "Ammonia is a weak base because it only partially ionizes in water to produce OH- ions."

All three parts correct and complete, matching the book's own wording.

Q.3: Limitations of Arrhenius theory; Bronsted-Lowry explanation 4 / 4

- (i) "Arrhenius theory has it's 2 limitations. (i) This applies to aqueous solutions. (ii) This does not explain why compounds like C02, S02 are acidic. There is no H in C02 and S02."
- (ii) First attempt scored through, partly legible ( "...HN03 [is an] acid because it donates... NH3 [is a] base... NH4N03" ), rewritten cleanly below:
- "In this equation, HN03 is an acid because it donates a proton (H+) ions to NH3. NH3 is a base bcuz it accepts the H+ proton from HN03, forming NH4."
- Equation: "NH3 + HN03 → NH4+ + N03- "

Two correct, distinct limitations, and HNO3/NH3 correctly identified as acid/base by proton donation: direction is right both times. The scored-through first attempt on (ii) is a partly legible near-duplicate of the same correct answer, not an error; the rewritten version below it is clean and complete.

Q.4: Bronsted-Lowry pair; ionization of two bases 4 / 4

(i) "HN03 + H2O → H3O + NO3" (no charges shown on the equation).

"HN03 is the bronsted-lowry acid bcuz it donates a proton (H)+ to H2O. H2O is the bronsted lowry base bcuz it accepts a H+ proton from HN03."

2/2

(ii)(a) "KOH → K+ + OH-" (single arrow) · (ii)(b) "NH4OH ⇌ NH4+ + OH-" (reversible arrow) ·

"weak base:- NH4OH, bcuz it only partially ionizes in water." 2/2

Acid correctly named as the proton donor HNO3 (never the product H3O+): this is the exact re-test of exam-23's Q.2(ii) miss, and it is fixed. Both arrow types (single for the strong base, reversible for the weak base) are correct, and the weak base is correctly identified with a correct reason.

Q.5: Neutralization equations 4 / 4

(i) "NH4OH + HN03 → NH4NO3 + H2O" 2/2

(ii) "H2SO4 + Mg(OH)2 → MgSO4 + 2H2O" 2/2

Both equations correctly balanced on the first clean line (in (ii) her original arrow is struck through and redrawn just above the line; the final, correct equation is unchanged). This is the exact re-test of exam-23's coefficient miss on Na2CO3+HCl and Q.8(i): fixed here on a different equation with the same 2-water pattern.

Q.6: Table 10.1 comparison 4 / 4

| Property              | Acid (Seemab) | Base (Seemab)                                                               |
|-----------------------|---------------|-----------------------------------------------------------------------------|
| Taste                 | Sour          | Bitter                                                                      |
| Effect on blue litmus | Turns red     | No effect                                                                   |
| Effect on red litmus  | No effect     | Turns red, "blue" written above with "red" scribbled over (self-correction) |
| Effect on skin        | Corrosive     | Corrosive                                                                   |

Judgement call: the base/red-litmus cell shows "turns red" with "blue" inserted above it and "red" scored through: a self-correction to the right answer, accepted in full (4/4). Range 3.5–4/4 if the self-correction were refused instead and the surviving word read literally as "red". All other seven cells are correct without qualification, including both rows correctly stated as corrosive per the book's own table.

Q.7: [Revision] ΔH for 2HI → H2 + I2 1 / 4

Formula: "ΔH°: Sum of bond dissociation energy of reactants – sum of formation energy of products" 1/1

Substitution: "ΔH° = [BFE of H-H + BFE of I-I] – (2×BFE of H-I) = (436+151) – 2(299) = 587 – 598" (she writes "BFE", bond formation/dissociation energy, throughout) 0/1

Value: "ΔH° = –11 KJ/mol" 0/1

Conclusion: "{ent [crossed out] enothermic reaction}" then

"Reason: The reaction is enothermic because it's ΔH° is negative, indicating that energy is released." ("enothermic" is her cursive spelling of exothermic, a missed "x") 0/1

This is exactly the trap the paper was built to test: the question deliberately reverses the book's own Example 8.1 (which computes H2 + I2 → 2HI and gets –11 kJ/mol), asking instead for 2HI → H2 + I2. She assigned the bond energies backward: treating H-H and I-I (the actual **products** of this reaction) as "reactants", and 2×H-I (the actual **reactant** bonds) as "products": which exactly reproduces the book's forward-reaction number, –11 kJ/mol, instead of the correct reversed answer, +11 kJ/mol. Her conclusion ("exothermic, energy released") is logically consistent with her own (wrong) negative value, so the concluding-sentence habit is present and working: the content is simply built on the wrong number. Only the formula statement, written correctly in words, earns credit. Judgement-call range: 1–2/4: a more generous reading could credit the substitution step for using the correct three bond energies even though the assignment is reversed (+1); 1/4 is applied here because the reversed assignment is the specific misconception this question targets, not a numerical slip.

SECTION C: Long Questions (15.5 / 16)

Q.8: Proton donors and proton acceptors 8 / 8

(i)(a) HCO3– + H2O ⇌ CO3^2– + H3O+: "HCO3- = an acid bcuz it donates H+ to H2O." "H2O = a base bcuz it accepts H+ from HCO3." 2/2

(i)(b) HCL + HCO3– ⇌ H2CO3 + Cl–: "HCL = an acid because it donates H+ to HCO3" "HCO3 = a base bcuz it accepts H+ from HCL." 2/2

(ii) "HCO3- is amphoteric bcuz it can act as both an acid and a base." "\* In (a), it acts as an acid by donating the protons." ·  
"· In (b), it acts as a base by accepting H+ [scored through: "from HC"]" 2/2

(iii) "Water is a proton acceptor (base) because it accepts H+ from HS-, forming H3O." 2/2

Every acid/base identification is correct, and crucially the acid is always named as the species donating the proton on the left of the arrow: never the product: across all four sub-parts. This is the strongest possible re-test result for exam-23's Q.2(ii) miss. Part (ii)'s first sentence ("donating the protons.") is complete on re-reading, not left blank; the second sentence is cut off mid-word by a scored-through "from HC", her own self-correction, not an omission.

Q.9: [Revision] Zinc and hydrochloric acid, 6.5 g Zn7.5 / 8

(i)"moles = 6.5g / 65g/mol = 0.1 mol"1/1

(ii)"1 mol Zn : 2 mol HC" then "0.1×2 = 0.2" (no unit on this line) "; molar mass HCl = 1+35.5 = 36.5 g/mol; mass = 0.2×36.5 = 7.3g"1.5/2

(iii)"1 mol Zn : 1 mol H2 → 0.1 mol H2; molar mass H2 = 2(1) = 2g/mol; mass = 0.1×2 = 0.2g"2/2

(iv)"number of molecules = moles × NA = 0.1×6.022×10^23 = 6.022×10^22 molecules"1/1

(v)"molar mass ZnCl2 = 65+2(35.5) = 136 g/mol; mass = 0.1×136 = 13.6g"1/1

(vi)"Therefore, 0.2g of hydrogen gas and 13.6g of zinc chloride are produced."1/1

An otherwise flawless stoichiometry chain, but in (ii) the intermediate mole result is written as a bare "0.2" with no "mol" unit: the paper's HIDEAAT #5 and the key's own note on this question both take ½ off for a missing unit, even though the final mass (7.3g) does carry its unit correctly. (i) and (iii)–(vi) are all unit-complete. Deep check 2026-09-22 corrected this from an initial 8/8 (which misquoted the ratio line as "0.1×2 = 0.2 mol"). **Judgement call:** a generous reading could treat the "2 mol" already written in the ratio line as carrying forward to the "0.2" result, restoring the full 2/2; 1.5/2 is applied as the stricter, literal reading. This is the strongest re-test result available for exam-23's Q.7 missing-conclusion miss, with one narrower unit slip surviving inside it.

RE-TEST OUTCOMES (exam-23's documented misses, deliberately re-served)

| Re-test                                                                                                                                              | Result    | Evidence                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MCQ 2, MCQ 4, Q.3(ii), Q.4(i), Q.8: acid is always the proton donor on the left of the arrow, never a product (exam-23 Q.2ii named H3O+ as the acid) | FIXED     | All five items correctly name the left-of-arrow species as the acid; H3O+ is never mistaken for the acid anywhere on this paper.                                                                                                                                                                                          |
| Q.4(ii): ionization direction and arrow choice (exam-23 Q.3)                                                                                         | FIXED     | KOH → K+ + OH− (single arrow, strong base) and NH4OH ⇌ NH4+ + OH− (reversible arrow, weak base) both correct.                                                                                                                                                                                                             |
| Q.5(ii): balancing, needs the coefficient on water (exam-23 Q.4ii, Q.8i)                                                                             | FIXED     | H2SO4 + Mg(OH)2 → MgSO4 + 2H2O, correctly balanced.                                                                                                                                                                                                                                                                       |
| Q.9: units and a full concluding sentence in a calculation (exam-23 Q.7)                                                                             | PARTIAL   | Mostly fixed: (i) and (iii)–(vi) all carry their unit and Q.9(vi) states both final masses in a complete sentence, but (ii)'s intermediate result "0.1×2 = 0.2" is left as a bare number, one unit still missing.                                                                                                         |
| Q.7: units and a concluding sentence in a calculation (exam-23 Q.7)                                                                                  | PARTIAL   | The sentence-writing habit held (a reasoned conclusion is present), but it states the wrong thing because the underlying value is wrong: see the row below.                                                                                                                                                               |
| MCQ 8: ΔH sign, positive = endothermic (exam-23 Q.9, ΔH never labelled)                                                                              | FIXED     | Answered B (endothermic) for ΔH°=+131.4kJ, correct.                                                                                                                                                                                                                                                                       |
| Q.7: ΔH sign, deliberately reversing the book's own Example 8.1 (exam-23 Q.9)                                                                        | NOT FIXED | Computed ΔH=−11kJ/mol (exothermic) by assigning the bond energies as if solving the book's forward reaction, reproducing its answer instead of the correct reversed one (+11kJ/mol, endothermic). Exactly the trap flagged in advance: "a student who copies −11 from the example loses the answer and conclusion marks." |

Of the 7 re-tests listed: 4 FIXED, 2 PARTIAL, 1 NOT FIXED. Both the Q.9 partial and the Q.7 partial/miss pair trace back to the same family of items (units-and-conclusion discipline, and the deliberately reversed ΔH sign trap), which together were the hardest, most deliberately-engineered material on the paper.

JUDGEMENT CALLS: range of defensible totals

| # | Item                                                                                                                   | Awarded | Range   | Note                                                                                                                                                                                                                                      |
|---|------------------------------------------------------------------------------------------------------------------------|---------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Q.9(ii): intermediate mole result "0.2" written with no unit                                                           | 1.5/2   | 1.5–2/2 | A generous reading could treat the "2 mol" already stated in the ratio line as carrying forward to "0.2"; 1.5/2 applied as the stricter, literal reading (a missing unit costs ½ per HIDEAAT #5 and the key's own note on this question). |
| 2 | Q.6: red-litmus/base cell: "turns red" with "blue" inserted above and "red" scored through                             | 4/4     | 3.5–4/4 | Accepted as a self-correction to the right answer; the stricter reading (refusing the self-correction, reading "red" literally) would cost ½ mark.                                                                                        |
| 3 | Q.7: formula statement correct; substitution, value and conclusion all built on a reversed reactant/product assignment | 1/4     | 1–2/4   | A generous reading could credit the substitution step for using the three correct bond energies despite the reversed assignment (+1); 1/4 applied because the reversal is the exact misconception being tested.                           |

Range for the record: 46/50 at the strictest reading (Q.9ii unit still off, Q.6 self-correction refused), up to 48/50 at the most generous reading (Q.9ii unit credited, Q.7 substitution credited). 46.5/50 is applied. Three items on this paper carry some ambiguity (Q.9ii, Q.6, Q.7); every other question matches the key without qualification.

STRENGTHS AND WEAKNESSES

**Strengths:** A perfect 10/10 on MCQs, independently re-derived from the paper. Chapter 10 (the brand-new content) scored a clean 35/35 (100%): up from 70.0% on exam-23's first attempt at this chapter. Every Bronsted-Lowry acid/base identification across the whole paper (Q.3ii, Q.4i, Q.8 all four parts) correctly names the proton donor on the left of the arrow as the acid, never the product: a complete fix of exam-23's Q.2(ii) miss. Both arrow conventions (single for strong, reversible for weak) are used correctly every time. Every equation in Q.5 is correctly balanced, including the second-water coefficient that exam-23 missed. Q.9's six-part stoichiometry chain is essentially flawless, with a full and correctly worded concluding sentence for both final masses.

**Weaknesses:** (1) **Reversed bond-energy assignment on a reversed reaction (Q.7):** when a reaction is written in reverse of a worked example she has seen, she reused the example's direction rather than re-deriving which bonds break and which form for the equation actually asked. This is a one-question miss but it is the same kind of "direction" error family as exam-23's ionization equations, now appearing in bond-energy substitution instead. (2) **One bare intermediate number without a unit (Q.9ii):** "0.1×2 = 0.2" has no "mol" label, even though the final answer on the same line (7.3g) is unit-complete; the units habit is mostly but not fully fixed.

COMPARISON: RECENT CHEMISTRY ATTEMPTS

| Exam          | Date       | Score | %     | MCQ  | Note                       |
|---------------|------------|-------|-------|------|----------------------------|
| exam-21-ch1-9 | 9 Sep 2026 | 41/46 | 89.1% | 7/12 | Written work perfect 16/16 |
|               |            |       |       |      |                            |

|                            |             |                |              |       |                                                                                                         |
|----------------------------|-------------|----------------|--------------|-------|---------------------------------------------------------------------------------------------------------|
| exam-22-ch1-9              | 14 Sep 2026 | 42/50          | 84.0%        | 9/10  | 6 of 9 re-set errors fixed                                                                              |
| exam-23-ch10-rev4-6-8-9    | 21 Sep 2026 | 36/50          | 72%          | 9/10  | First-ever Ch 10 paper; Ch 10 alone 24.5/35 (70.0%)                                                     |
| <b>exam-25-ch10-rev6-8</b> | 22 Sep 2026 | <b>46.5/50</b> | <b>93.0%</b> | 10/10 | Second Ch 10 paper; Ch 10 alone <b>35/35 (100%)</b> ; best Chemistry score since exam-17 (3 Aug, 96.9%) |

A 21-point rebound from exam-23. The Chapter 10 portion specifically went from 70.0% to a clean 100% on its second outing: every one of exam-23's targeted re-tests on this chapter (proton-donor identification, arrow choice, balancing) is now fixed. The only miss on the whole paper is the single hardest item deliberately built into it: correctly re-deriving direction when the book's own worked example is reversed.

End of Report